

Hierarchical Supervisory Control of Discrete Event Systems: Normality Under Masks

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Abstract— In this paper, we study hierarchical supervisory control of discrete event systems under mask-based partial observation. Unlike natural projection, a mask may map distinct observable events to the same observation symbol, and such a symbol does not need to belong to the original event alphabet. This more general observation model captures indistinguishable sensing and observation corruption. However, extending hierarchical synthesis to the mask-based setting is nontrivial, since mask and abstraction do not, in general, commute; yet, commutativity is a property that has been heavily used for projection-based observations. Consequently, some trivial properties from projection-based observations, such as the relation between the indistinguishability of two strings and that of the corresponding abstracted strings, no longer hold for masks. To ensure hierarchical consistency under masks and preserve normality between the low-level and high-level plant models, forward mask consistency and backward mask consistency are introduced as sufficient conditions. Finally, the strong backward mask consistency condition is used to guarantee that supremal normal sublanguages are preserved between both levels. Examples are presented to illustrate the proposed method.

Index Terms—Discrete event system, hierarchical supervisory control, observation consistency, normality.

I. INTRODUCTION

DISCRETE event systems (DESs) [1] play an important role in the modeling and analysis of computer-extensive systems, particularly in engineering fields such as flexible manufacturing and transportation systems. With the continuous increase in system size and operational complexity, classical control paradigms often encounter significant computational challenges [2], necessitating the adoption of hierarchical and modular control architectures [3]–[4]. Hierarchical supervisory control of DESs was first introduced by Zhong and Wonham in 1990 [5], presented as a two-level vertically decomposed system. The setup embodies a low-level detailed model and a high-level abstracted model controlled by a low-level supervisor and a high-level supervisor, respectively. The two levels are connected through command and information channels.

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The objective is to synthesize a non-blocking and maximally permissive high-level supervisor to achieve the desired low-level implementation, referred to as hierarchical consistency.

In traditional hierarchical supervisory control of DESs, abstraction is typically achieved through natural projections that select a subset of the original event alphabet for control [5]. However, in many practical systems, several different observations can be grouped into a category. For example, in automated manufacturing systems, multiple isomorphic components such as automated guided vehicles, workstations, or buffers may be only partially distinguishable to the supervisor because of limited sensing resolution or aggregated communication interfaces. Motivated by this observation, this paper investigates partially observable DESs, employing a mask-based observation mechanism, as presented in [6]–[7]. A mask can map multiple observable events of a plant to the same abstract label (which may not belong to the original system alphabet), while still allowing event unobservability.

To the best of our knowledge, hierarchical supervisory control with normality under mask-based observation has not yet been studied. To place this gap in context, we briefly review the development of hierarchical control of DESs. In the fully observable setting, a sufficient condition of output control consistency to guarantee hierarchical consistency is studied in [5]. An abstract hierarchical control theory is developed in [8] by establishing control consistency and observer property based on aggregation mapping in a two-level hierarchy. The computational complexity of control consistency in [8] is reduced in [9]. In [10], a less restrictive sufficient condition of local control consistency is formulated for maximally permissive control. However, the above approaches assume full observability of a plant.

Under partial observation, hierarchical control requires additional conditions to ensure that observability is preserved between the two levels [12]. A feasible low-level supervisor is considered in [11] to implement the high-level control commands. The study in [12] proposes a sufficient condition, called observation consistency, which allows hierarchical synthesis on the high-level (abstracted) plant and addresses a certain consistency property regarding the observations of strings on both high-level and low-level structures. A stronger condition, called modified observation consistency, is defined in [13]–[14], under which the supremal normal sublanguages of specifications between the levels coincide.

It is worth noting that extending hierarchical control from projection-based observations to mask-based ones introduces several challenges, since some fundamental properties that hold for projections do not necessarily hold for masks. For

instance, the indistinguishability between the events in the original system defined by the mask does not necessarily imply indistinguishability in the abstracted system. Another motivation for studying masks, beyond allowing indistinguishability among observable events, lies in their relevance to cybersecurity, where masks may represent simple cyber-attacks in which observable events can be not only deleted but also replaced by an attacker. To this end, this paper focuses on the computation of abstraction-based supervisors corresponding to supremal normal sublanguages. The main contributions are as follows.

- This paper establishes a forward mask consistency (FMC) for DESs under observation masks to preserve indistinguishability, motivated by the fact that, unlike projection-based observations, indistinguishability by the masks of two strings over the original alphabet does not imply indistinguishability of the corresponding abstracted strings over the high-level alphabet in hierarchical supervisory control.
- To ensure FMC, we develop a new condition, under which either all indistinguishable events are in the high-level or none of them is in the high level. Importantly, this condition does not require the commutativity of mask-based partial observations with abstractions, yet still guarantees FMC.
- It is shown that supremal normal supervisors computed on the high-level are also maximally permissive when implemented on the low-level under FMC and a strong backward mask consistency.

The paper is organized as follows. The basics of supervisory control are introduced in Section II. Section III presents the hierarchical control with observation masks. In Section IV, we propose the main results for implementing hierarchical control under masks and illustrate them through an example. Section V extends the results to the supremal controllable and normal sublanguages. Finally, Section VI concludes this paper.

II. PRELIMINARIES

We assume that the reader is familiar with the preliminaries of supervisory control [1]. Let Σ be a finite nonempty set of events and Σ^* be the set of all finite strings over Σ , including the empty string ε . Given a string $w \in \Sigma^*$, its length is denoted by $|w|$ (the length of the empty string is defined as $|\varepsilon| = 0$). A language $L \subseteq \Sigma^*$ is a set of finite-length strings. A prefix of a string w is a string $u \in \Sigma^*$ for which there exists a string $v \in \Sigma^*$ such that $uv = w$, where uv denotes the concatenation of strings u and v . The prefix closure of w is the set of all its prefixes, denoted by $\bar{w} = \{u \in \Sigma^* \mid \exists v \in \Sigma^*: uv = w\}$. The prefix closure of a language L is the set $\bar{L} = \{w' \in \Sigma^* \mid \exists w \in L : w' \in \bar{w}\}$. A language L is prefix-closed if $L = \bar{L}$.

A *projection* $P : \Sigma^* \rightarrow \Gamma^*$ with $\Gamma \subseteq \Sigma$, is defined as follows: $P(\varepsilon) = \varepsilon$; $P(\sigma) = \varepsilon$ if $\sigma \in \Sigma \setminus \Gamma$; $P(\sigma) = \sigma$ if $\sigma \in \Gamma$; $P(s\sigma) = P(s)P(\sigma)$ for $s \in \Sigma^*$ and $\sigma \in \Sigma$. The action of P on a string $\sigma_1\sigma_2 \cdots \sigma_n$ is to remove the events that are not in Γ . Such a projection is usually called *natural projection*. The inverse image of a string $w \in \Gamma^*$ under the projection P is the set $P^{-1}(w) = \{s \in \Sigma^* \mid P(s) = w\}$. The definitions can naturally be extended to languages.

A *deterministic finite automaton* (DFA) is a quintuple $G = (X, \Sigma, \delta, x_0, X_m)$, where X is the finite set of states; Σ is the finite set of events; $x_0 \in X$ is the initial state; $X_m \subseteq X$ is the set of marked states; $\delta : X \times \Sigma \rightarrow X$ is the transition function that can be extended to the domain $X \times \Sigma^*$ in the usual way. We say that $L(G) = \{w \in \Sigma^* \mid \delta(x_0, w) \in X_m\}$ is the generated language of G , while $L_m(G) = \{w \in \Sigma^* \mid \delta(x_0, w) \in X_m\}$ is the marked or accepted language of G . By definition, $L_m(G) \subseteq L(G)$, and $\bar{L(G)}$ is prefix-closed. A DFA G is said to be non-blocking if $\bar{L_m(G)} = L(G)$.

Given a DES G and a set of mask labels (another alphabet) Δ , a *mask* $M : \Sigma \rightarrow \Delta \cup \{\varepsilon\}$ is a function that assigns to an event in Σ either a symbol in Δ or the empty string ε . Furthermore, the mask M can be extended to strings, defined as $M : \Sigma^* \rightarrow \Delta^*$, that is, $M(se) = M(s)M(e)$ and $M(\varepsilon) = \varepsilon$, where $e \in \Sigma$ and $s \in \Sigma^*$. The inverse image of a string $w \in \Delta^*$ under the mask M is the set $M^{-1}(w) = \{s \in \Sigma^* \mid M(s) = w\}$. The definitions can be readily extended to languages.

Given a DES G , the set of events Σ is partitioned into two disjoint subsets, i.e., $\Sigma = \Sigma_o \dot{\cup} \Sigma_{uo}$, where Σ_o is the set of observable events and Σ_{uo} is that of unobservable events. From the perspective of controllability, Σ can be divided into the set of controllable events Σ_c and that of uncontrollable events Σ_{uc} , i.e., $\Sigma = \Sigma_c \dot{\cup} \Sigma_{uc}$. A language $K \subseteq L(G)$ is *controllable* with respect to $L(G)$ and the set Σ_{uc} of uncontrollable events if $\bar{K}\Sigma_{uc} \cap L(G) \subseteq \bar{K}$. A language K is *normal* with respect to $L(G)$ and the mask $M : \Sigma \rightarrow \Delta \cup \{\varepsilon\}$ if for all $s \in \bar{K}$ and $s' \in L(G)$ such that $M(s') = M(s)$, $s' \in \bar{K}$ holds. The supremal normal sublanguage of the language K with respect to $L(G)$ and the mask M is denoted by $\text{supN}(K, L(G), M)$.

The parallel composition of languages $L_i \subseteq \Sigma_i^*$ is the language $\|_{i=1}^n L_i = \cap_{i=1}^n P_i^{-1}(L_i)$, where $P_i : (\cup_{i=1}^n \Sigma_i)^* \rightarrow \Sigma_i^*$ is the projection, for $i = 1, \dots, n$ [1]. In particular, for DFAs G_i , we have $L(\|_{i=1}^n G_i) = \cap_{i=1}^n L(G_i)$ and $L_m(\|_{i=1}^n G_i) = \cap_{i=1}^n L_m(G_i)$. The languages L_i are (synchronously) non-conflicting if $\|_{i=1}^n \bar{L}_i = \bar{\|_{i=1}^n L_i}$.

Given a DES G over Σ and a natural projection $P : \Sigma^* \rightarrow \Sigma_o^*$, we denote the set of control patterns by $\Gamma = \{\gamma \subseteq \Sigma \mid \Sigma_{uc} \subseteq \gamma\}$. A *supervisor* of G with respect to a set of control patterns Γ is a map $S : P(L(G)) \rightarrow \Gamma$ returning, for every observed string, a set of enabled events that always include all uncontrollable events. The *closed-loop system* of G under the supervision of S is the minimal language $L(S/G)$ such that $\varepsilon \in L(S/G)$ and $sa \in L(S/G)$ iff $s \in L(S/G)$, $sa \in L(G)$, and $a \in S(P(s))$. If the closed-loop system is non-blocking, i.e., $\bar{L_m(S/G)} = L(S/G)$, then S is said to be non-blocking. Intuitively, a supervisor disables some of the controllable transitions according to the partial observation of the system.

III. HIERARCHICAL SUPERVISORY CONTROL

Before stating the hierarchical supervisory control problem for partially observable DESs under observation masks, we review the notations of hierarchical supervisory control [3]. Given the set of events Σ and the set of mask labels Δ , we

denote the high-level alphabet by $\Sigma_{hi} \subseteq \Sigma$, and the set of high-level mask labels by $\Delta_{hi} \subseteq \Delta$. The system's observation is described by a mask $M : \Sigma^* \rightarrow \Delta^*$. Let projection $Q : \Sigma^* \rightarrow \Sigma_{hi}^*$ be the high-level abstraction. In this way, the high-level observation is defined as the restriction of M to Σ_{hi} , i.e., $M_{hi} : \Sigma_{hi}^* \rightarrow \Delta_{hi}^*$. The abstraction on the set of mask labels Δ is given by the projection $Q_\Delta : \Delta^* \rightarrow \Delta_{hi}^*$. The relationships among M , Q , M_{hi} and Q_Δ are illustrated in Fig. 1. Note that in general mask M with abstraction Q may not commute, e.g., a low-level event $a \in \Sigma \setminus \Sigma_{hi}$ is abstracted away by Q , i.e., $M_{hi}(Q(a)) = \varepsilon$, but it is possible that $M(a) = m \in \Delta$ and $Q_\Delta(M(a)) = m \in \Delta_{hi}$.

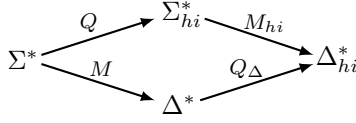


Fig. 1. Diagram of abstractions and masks.

Problem 1 (Hierarchical supervisory control). Let G be a low-level plant over Σ , and K be a high-level specification over $\Sigma_{hi} \subseteq \Sigma$. The high-level plant G_{hi} over Σ_{hi} is defined as $L(G_{hi}) = Q(L(G))$ and $L_m(G_{hi}) = Q(L_m(G))$, where Q is an abstraction. Given G_{hi} and K , find a non-blocking low-level supervisor S such that $L_m(S/G) = K \parallel L_m(G)$. $\diamond$

Intuitively, we need to design the projection Q , masks M and M_{hi} such that the controllability and normality of $K \parallel L_m(G)$ for the low-level plant G are equivalent to K for the high-level plant G_{hi} . It is proposed in [12] that observation consistency serves as a sufficient condition for establishing such equivalence, while no condition is stated to guarantee maximal permissiveness. Moreover, observation consistency is not usable when observable events are masked, and an additional condition is required. It is necessary to explore a new condition to preserve the consistency. Now, we present the definition of observation consistency under a mask condition.

Definition 1 (Backward mask consistency (BMC)). A prefix-closed language $L \subseteq \Sigma^*$ is backward mask consistent w.r.t. Q , M , and M_{hi} if for every pair of high-level strings $t, t' \in Q(L)$ with $M_{hi}(t) = M_{hi}(t')$, there exist strings $s, s' \in L$ such that $Q(s) = t$, $Q(s') = t'$, and $M(s) = M(s')$. $\square$

Definition 1 requires that any two strings with the same observation mask in the abstracted language $Q(L)$ should have corresponding strings in the original plant with the same observation mask. Although the verification of BMC is PSPACE-hard and its decidability remains an open question [14], we will present a verifiable sufficient condition in Proposition 10 to satisfy BMC.

Under the natural projection, observational indistinguishability (projections equivalence) at the low level implies that at the high level, since the high-level alphabet is a subset of the original alphabet. However, this implication does not hold under mask-based observations. The reason for this discrepancy is that for observation masks, the commutativity between masks and abstraction (natural projection) may not

hold, while for standard natural projection observation, the two natural projections always commute. To this end, a forward mask consistency is proposed.

Definition 2 (Forward mask consistency (FMC)). A prefix-closed language $L \subseteq \Sigma^*$ is said to be forward mask consistent with respect to Q , M , and M_{hi} if for all strings $s, s' \in L$ with $M(s) = M(s')$, it holds $M_{hi}(Q(s)) = M_{hi}(Q(s'))$. $\square$

Remark 3. Note that FMC requires that any two strings with the same observation mask in the low-level plant should have abstracted strings with the same observation mask in the high-level plant, as shown in Fig. 2. It is worth noting that two events (or strings) with the same mask may be abstracted into the events (or strings) that have different masks on the high-level alphabet. However, it should be clear that the commutativity in Fig. 1 implies FMC, i.e., if $M_{hi}Q = Q_\Delta M$, then for any two strings s, s' with $M(s) = M(s')$, we have $M_{hi}(Q(s)) = Q_\Delta(M(s)) = Q_\Delta(M(s')) = M_{hi}(Q(s'))$. $\diamond$

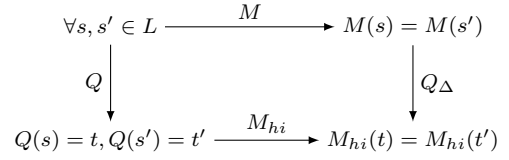


Fig. 2. Illustration of the FMC.

Note that there exists a weaker condition than normality, known as observability in supervisory control under partial observations [1]. However, there are two issues: (i) Language observability requires additional conditions compared with normality to be applied in the hierarchical setting [12]. (ii) Unlike normality, observability is not closed under union, suggesting that, given a set of observable sublanguages, the supremal element does not exist in general. Moreover, computing supremal relatively observable sublanguages is more computationally demanding than computing supremal normal sublanguages. Normality is preserved under union and implies observability. In particular, it becomes equivalent to observability when all controllable events are observable [1].

In the following, we establish that FMC and BMC jointly indicate the preservation of normality for the two-level languages. It allows to use a high-level supervisor that enforces the high-level specification K for the abstraction G_{hi} , which can be implemented for the original plant G .

Theorem 4. Let $G = (X, \Sigma, \delta, x_0, X_m)$ be a non-blocking generator and $K \subseteq Q(L_m(G))$ be a high-level specification. Assume that $L(G)$ is forward mask consistent and backward mask consistent w.r.t. Q , M , and M_{hi} , and that K and $L_m(G)$ are synchronously non-conflicting. Then the language K is normal with respect to $Q(L(G))$ and M_{hi} if and only if $K \parallel L_m(G)$ is normal with respect to $L(G)$ and M .

Proof: ($\Leftarrow$) For all strings $t' \in \bar{K} \subseteq Q(L)$ and for all $t \in Q(L)$ such that $M_{hi}(t') = M_{hi}(t)$, we need to show $t \in \bar{K}$. Since $t, t' \in Q(L)$ and $M_{hi}(t) = M_{hi}(t')$, by BMC, there exist $s, s' \in L$ such that $Q(s) = t$, $Q(s') = t'$, and $M(s) = M(s')$. Thanks to $t' \in \bar{K} \subseteq Q(L)$, we have $s' \in$

$Q^{-1}(\bar{K}) \cap L$, which means $s' \in \bar{K} \parallel L$. By the normality of $K \parallel L_m$ w.r.t. L and M , we have $s \in Q^{-1}(\bar{K}) \cap L$ and thus $t = Q(s) \in \bar{K} \cap Q(L) \subseteq K$, which completes the “if” direction.

($\Rightarrow$) Assume $s' \in \bar{K} \parallel L_m$ and $s \in L$ such that $M(s) = M(s')$. By FMC, for $t = Q(s)$ and $t' = Q(s')$, it follows that $M_{hi}(t) = M_{hi}(t')$. Thanks to the non-conflictingness assumption and $s' \in \bar{K} \parallel L_m$, we have $s' \in Q^{-1}(\bar{K}) \cap L$ and thus $t' = Q(s') \in \bar{K} \cap Q(L) \subseteq \bar{K}$. The normality of K w.r.t. $Q(L)$ and M_{hi} implies $Q(s) = t \in \bar{K}$. It holds $s \in Q^{-1}(\bar{K}) \cap L_m = \bar{K} \parallel L_m$, which concludes the claim. ■

Example 5. Consider the plant G in Fig. 3 with $\Sigma = \{a, b, c, d, e, f\}$ and high-level alphabet $\Sigma_{hi} = \{a, b, e, f\}$. The abstracted plant with the language $Q(L)$ and the specification K are depicted in Fig. 3 as well. Let the mask M be $M(a) = M(b) = M(c) = M(d) = m$, $M(e) = e$, and $M(f) = f$. Since every pair of strings with the same observation, such as $s = ac$ and $s' = bd$, is abstracted into the strings with the same high-level observation, the language L is forward mask consistent w.r.t. Q , M , and M_{hi} due to Definition 2. Since the indistinguishable high-level strings a and b have low-level strings ac and bd with the same observation, L is backward mask consistent w.r.t. Q , M , and M_{hi} . By Theorem 4, K is normal w.r.t. $Q(L)$ and M_{hi} if and only if $K \parallel L_m$ is normal w.r.t. L and M . ■

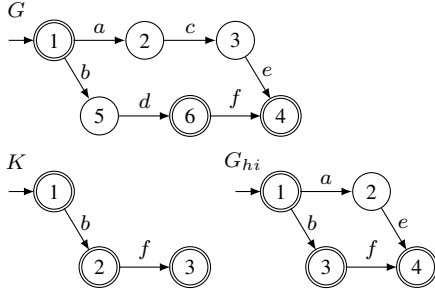


Fig. 3. Plant G , specification K , and abstraction G_{hi} .

As shown in Example 5, ensuring FMC requires the identification of every pair of strings that look the same in the language of the low-level plant, which is computationally expensive. To this end, we propose an easily checkable condition to ensure FMC.

Proposition 6. Let G be a DES over Σ . If for all $\sigma \in \Sigma_{hi}$, for all $\sigma' \in \Sigma$ such that $M(\sigma') = M(\sigma) \neq \varepsilon$, $\sigma' \in \Sigma_{hi}$ holds, then $L(G)$ is forward mask consistent.

Proof: The proof is conducted by structural induction along the strings $M(s) = M(s') \in \Delta^*$. The base case $M(s) = M(s') = \varepsilon$ is trivial. Making the induction hypothesis that $M(\omega) = M(\omega') = w \in \Delta^*$ implies $M_{hi}Q(\omega) = M_{hi}Q(\omega')$, it remains to show that for $M(s) = M(s') = w\delta \in \Delta^*$, $M_{hi}Q(s) = M_{hi}Q(s')$. For $M(s) = M(s') = w\delta \in \Delta^*$, there exist decompositions $s = \omega u$ and $s' = \omega' u'$, where $M(\omega) = M(\omega') = w$ and $u = u_1 \dots u_n, u' = u'_1 \dots u'_m \in \Sigma^*$ are strings such that $M(u) = M(u') = \delta \in \Delta$. By the definition of mask, there exist $k \in \{1, \dots, n\}$ and $k' \in \{1, \dots, m\}$ such that $M(u_k) = \delta$, $M(u_j) = \varepsilon$ for $j \neq k$

and $M(u'_{k'}) = \delta$, $M(u'_i) = \varepsilon$ for $i \neq k'$. By the assumption that for all $\sigma \in \Sigma_{hi}$ and $\sigma' \in \Sigma$, $M(\sigma') = M(\sigma) \neq \varepsilon$ implies $\sigma' \in \Sigma_{hi}$, we conclude two possible cases by $M(u_k) = M(u'_{k'}) = \delta \neq \varepsilon$: (i) $u_k, u'_{k'} \in \Sigma_{hi}$, with $M_{hi}Q(u_k) = M_{hi}(u_k) = \delta = M_{hi}(u'_{k'}) = M_{hi}Q(u'_{k'})$ and (ii) $u_k, u'_{k'} \notin \Sigma_{hi}$, with $M_{hi}Q(u_k) = \varepsilon = M_{hi}Q(u'_{k'})$. Finally, concerning other u'_i s and u'_j s, $i \neq k, j \neq k'$, we have $M_{hi}Q(u_i) = \varepsilon = M_{hi}Q(u'_{k'})$, due to either $u_i \notin \Sigma_{hi}$, i.e., $M_{hi}Q(u_i) = M_{hi}(\varepsilon) = \varepsilon$ or $u_i \in \Sigma_{hi}$, in latter case $M_{hi}Q(u_i) = M_{hi}(u_i) = \varepsilon$ thanks to $M(u_i) = \varepsilon$ for $i \neq k$. The same arguments apply to u'_j for $j \neq k'$. Thus, $M_{hi}Q(\omega u) = M_{hi}Q(\omega u')$ and $M_{hi}Q(s) = M_{hi}Q(\omega u) = M_{hi}Q(\omega u') = M_{hi}Q(s')$, which completes the proof. ■

It should be noted that the abstraction consistency is only required for observable parts of indistinguishable events, i.e., their masks correspond to some observations in $\delta \in \Delta$. Therefore, the sufficient condition for FMC presented in Proposition 6 is weaker than the condition $M^{-1}M(\Sigma_{hi}) \subseteq \Sigma_{hi}$. The latter is also equivalent to $M^{-1}M(\Sigma \setminus \Sigma_{hi}) \subseteq \Sigma \setminus \Sigma_{hi}$. However, the condition in Proposition 6 is only sufficient but not necessary for FMC. Considering the same DES G as in Example 5, L is forward mask consistent. We have $M(a) = M(c) = m \neq \varepsilon$, where $a \in \Sigma_{hi}$ but $c \notin \Sigma_{hi}$. This violates the condition in Proposition 6.

On the other hand, it turns out that the sufficient condition in Proposition 6 is weaker than another sufficient condition for FMC, namely the diagram commutativity in Fig. 1, which is proved in Remark 3. This leads to the following result.

Proposition 7. For a DFA G over Σ , if $M_{hi}Q = Q_{\Delta}M$ holds, then for all $\sigma \in \Sigma_{hi}$ and for all $\sigma' \in \Sigma$ such that $M(\sigma') = M(\sigma) \neq \varepsilon$, it follows that $\sigma' \in \Sigma_{hi}$.

Proof: Suppose that $M_{hi}Q = Q_{\Delta}M$ holds. Then for all $\sigma \in \Sigma_{hi}$, $\sigma' \in \Sigma$ such that $M(\sigma) = M(\sigma') \neq \varepsilon$, we have $M_{hi}Q(\sigma) = Q_{\Delta}M(\sigma) = Q_{\Delta}M(\sigma') = M_{hi}Q(\sigma')$. By $\sigma \in \Sigma_{hi}$, it follows that $Q(\sigma) = \sigma \neq \varepsilon$. Then $M_{hi}Q(\sigma) = M_{hi}(\sigma) = M_{hi}Q(\sigma')$. Assume by contradiction that $\sigma' \in \Sigma \setminus \Sigma_{hi}$, i.e., $Q(\sigma') = \varepsilon$. Then $M_{hi}(\sigma) = M_{hi}Q(\sigma') = M_{hi}(\varepsilon) = \varepsilon$, which contradicts the assumption $M(\sigma) = M(\sigma') \neq \varepsilon$, i.e., $\sigma' \in \Sigma_{hi}$. ■

IV. SUPREMACY OF NORMAL SUPERVISORS

The hierarchical supervisory control problem requires the computation of the supervisor based only on the abstracted system. In the previous section, the conditions for preserving the normality are discussed in Theorem 4. Although a high-level supervisor can be implemented on the original system, the supervisor is generally not maximally permissive. To address this issue, we first define the maximal permissiveness and then elaborate on the problem.

Definition 8 (Maximally permissive supervisor). Given a DFA G and a high-level specification K , a supervisor S is said to be maximally permissive if for any supervisor S' such that $L_m(S'/G) \subseteq K \parallel L_m(G)$, $L_m(S'/G) \subseteq L_m(S/G)$ holds. □

Note that in the partially observable DESs, multiple incomparable solutions may exist. However, if all controllable

events are observable, then controllability and observability jointly imply normality [1], which guarantees the existence of a supremal solution.

Theorem 4 states that BMC and FMC imply the normality preservation of the abstraction-based supervisor implemented on the low-level system. However, these conditions do not ensure maximal permissiveness of the closed-loop system. Specifically, a high-level supervisor based on the supremal normal (and controllable) sublanguage with respect to the abstracted plant may be more restrictive when implemented on the original plant than the supremal normal (and controllable) sublanguage computed directly for the original plant.

Under the same setting as in Problem 1, the hierarchical supervisory control synthesis problem aims to find a low-level supervisor S that is maximally permissive (Definition 8) and non-blocking, i.e., $L_m(S/G) \subseteq K \parallel L_m(G)$, utilizing the high-level plant G_{hi} and the high-level specification K .

Problem 2 (Maximal permissiveness problem of hierarchical supervisory control). Consider a low-level plant G and a high-level specification $K \subseteq Q(L_m(G))$, where Q is an abstraction. Find the conditions under which the low-level supremal normal (and controllable) sublanguage equals the high-level supremal normal (and controllable) sublanguage implemented on the original system, i.e., $\sup N(K \parallel L_m, L, M) = \sup N(K, Q(L), M_{hi}) \parallel L_m$, where $L = L(G)$ and $L_m = L_m(G)$. $\diamond$

Intuitively, to avoid the direct computation of $\sup N(K \parallel L_m, L, M)$, we instead compute $\sup N(K, Q(L), M_{hi})$ on the high-level plant and then derive its implementation on the low-level plant. The closed-loop behavior on the low-level plant is $\sup N(K, Q(L), M_{hi}) \parallel L_m = Q^{-1}(\sup N(K, Q(L), M_{hi})) \cap L_m$, where L_m is the marked behavior of the low-level plant.

Unfortunately, BMC and FMC are not sufficient to guarantee that $\sup N(K, Q(L), M_{hi}) \parallel L_m$ is equal to $\sup N(K \parallel L_m, L, M)$. To find the supremal normal sublanguages under projection-based observations, Komenda et al. [14] introduce a new condition called modified observation consistency. Now, we need to extend this condition to cope with mask-based observations.

Definition 9 (Strong backward mask consistency (SBMC)). A prefix-closed language $L \subseteq \Sigma^*$ is said to be strong backward mask consistent with respect to projection Q , masks M , and M_{hi} if for all $s \in L$, and $t' \in Q(L)$ with $M_{hi}(Q(s)) = M_{hi}(t')$, there is a string $s' \in L$ such that $Q(s') = t'$ and $M(s) = M(s')$. $\square$

Definition 9 indicates that every string of the high-level plant that looks the same as a low-level string s has a corresponding string s' that looks the same as s , as shown in Fig. 4. We will not provide an algorithm to check SBMC, but rather present a checkable condition. For this purpose, for a DFA G over the event set Σ , let $\Sigma_{om} = \{\sigma \in \Sigma \mid M(\sigma) \neq \varepsilon\}$ be the set of mask-based observable events.

Proposition 10. Let G be a DES over Σ , and let Σ_{om} and Σ_{hi} be the mask-based observable and high-level alphabets,

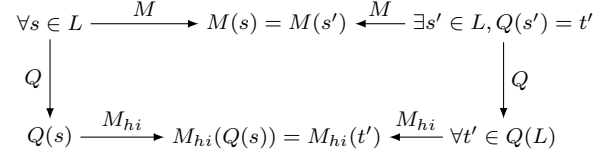


Fig. 4. Illustration of SBMC.

respectively. If $\Sigma_{om} \subseteq \Sigma_{hi}$, then $L(G)$ is strong backward mask consistent.

Proof: We first show that if $\Sigma_{om} \subseteq \Sigma_{hi}$, then $M = M_{hi}Q$. To see this formally, we consider the equality on events, from which the equality on strings (and languages) follows by induction and the property of masks and projections. Let us consider two cases. For $\sigma \in \Sigma_{om}$, we have $\sigma \in \Sigma_{hi}$, i.e., $Q(\sigma) = \sigma$, implying $M(\sigma) = M_{hi}Q(\sigma)$. For $\sigma \notin \Sigma_{om}$, i.e., $M(\sigma) = \varepsilon$, we have two cases: either $\sigma \notin \Sigma_{hi}$, $Q(\sigma) = \varepsilon$, i.e., $M(\sigma) = \varepsilon = M_{hi}(\varepsilon) = M_{hi}Q(\sigma)$, or $\sigma \in \Sigma_{hi}$, $Q(\sigma) = \sigma$, i.e., $M(\sigma) = \varepsilon = M_{hi}(\sigma) = M_{hi}Q(\sigma)$.

To see that SBMC holds, consider $s \in L$ and $t' \in Q(L)$ satisfying $M_{hi}(Q(s)) = M_{hi}(t')$. It suffices to choose any $s' \in L$ such that $Q(s') = t'$, which exists due to $t' \in Q(L)$. We will show that any $s' \in L$ with $Q(s') = t'$ verifies $M(s') = M(s)$. We have $M(s) = M_{hi}(Q(s)) = M_{hi}(t') = M_{hi}(Q(s')) = M(s')$, showing that L satisfies SBMC. $\blacksquare$

Note that SBMC implies BMC, i.e., any language L that is strong backward mask consistent also satisfies BMC, suggesting that $\Sigma_{om} \subseteq \Sigma_{hi}$ also guarantees that L is backward mask consistent. We next present an alternative sufficient condition for SBMC.

Proposition 11. Let G be a DES over Σ . If (i) $\Sigma_{hi} \subseteq \Sigma_{om}$ and (ii) for all $\sigma, \sigma' \in (\Sigma_o \cap \Sigma_{hi})^*$ with $\sigma \neq \sigma'$ implying $M_{hi}(\sigma) \neq M_{hi}(\sigma')$, then $L(G)$ is strong backward mask consistent.

Proof: Let $s \in L$ and $t' \in Q(L)$ such that $M_{hi}(Q(s)) = M_{hi}(t')$. According to (ii), M_{hi} is an injective map, i.e., $M_{hi}(Q(s)) = M_{hi}(t')$ implies $Q(s) = t'$. In this case, we can take $s' = s$ and complete the proof. $\blacksquare$

Under the alternative conditions given in Proposition 11, the high-level system has full observations, suggesting that supervisors can be computed in polynomial time. Propositions 10 and 11 provide two conditions for SBMC. These conditions are sufficient but not necessary, as illustrated in the following.

Example 12. Consider the alphabet $\Sigma = \{a, b, c\}$ with masks $M(a) = M(b) = m, M(c) = c$, high-level events $\Sigma_{hi} = \{a, b\}$, and language $L = \{\varepsilon, a, b, c\}$. Then, $\Sigma_{om} = \{a, b, c\}$ implies $\Sigma_{om} \not\subseteq \Sigma_{hi}$, i.e., the condition of Proposition 10 is violated. Since $M_{hi}(a) = M_{hi}(b) = m$, condition (ii) of Proposition 11 is violated. To check that SBMC holds, consider $Q(L) = \{\varepsilon, a, b\}$. For any $t' \in Q(L)$, there is $s \in L$ such that $M_{hi}Q(s) = M_{hi}(t')$. Namely, $s = Q(s) = t'$ holds for $t' = a$ or $t' = b$. Then, we can choose $s' = t'$. Similarly, for $t' = \varepsilon$, there is either $s = c$ or $s = \varepsilon$ to make $M_{hi}Q(s) = M_{hi}(\varepsilon)$. In both cases, we take $s' = s$. Then $Q(s') = t'$ and $M(s') = M(s)$ and SBMC of L holds, even though neither Proposition 10 nor 11 applies. $\blacksquare$

Now, we extend a result from [14], that is, SBMC guarantees that normality is preserved by abstractions, which is essential in the main result concerning the preservation of supremal normal supervisors.

Lemma 13. Given a DFA G with $L = L(G)$ and $L_m = L_m(G)$, if L is strong backward mask consistent with respect to Q , M , and M_{hi} and $K \subseteq L_m$ is normal with respect to L and M , then $Q(K)$ is normal with respect to $Q(L)$ and M_{hi} .

Proof: Since the inclusion $Q(\bar{K}) \subseteq M_{hi}^{-1}M_{hi}(Q(\bar{K})) \cap Q(L)$ is trivial, it suffices to show $M_{hi}^{-1}M_{hi}(Q(\bar{K})) \cap Q(L) \subseteq Q(\bar{K})$. Let $t' \in M_{hi}^{-1}M_{hi}(Q(\bar{K})) \cap Q(L)$. Then, there exists $s \in \bar{K}$ such that $M_{hi}(Q(s)) = M_{hi}(t')$. By SBMC, there is a string $s' \in L$ such that $Q(s') = t'$ and $M(s) = M(s')$, that is, $s' \in M^{-1}M(s) \cap L \subseteq M^{-1}M(\bar{K}) \cap L \subseteq \bar{K}$, where the last inclusion follows from the normality of K with respect to L and M . Hence, $t' = Q(s') \in Q(\bar{K})$ is true. ■

Now we show that, under FMC and SBMC, the high-level supervisor S_{hi} , for the high-level plant G_{hi} and specification K , can serve as a maximally permissive supervisor when implemented for the low-level plant.

Theorem 14. Let G be a non-blocking DFA with $L = L(G)$ and $L_m = L_m(G)$. For any high-level specification $K \subseteq Q(L_m)$,

$$\sup N(K \parallel L_m, L, M) = \sup N(K, Q(L), M_{hi}) \parallel L_m$$

if 1) $\sup N(K, Q(L), M_{hi})$ and L_m are non-conflicting, and 2) L is forward mask consistent and strong backward mask consistent with respect to Q , M , and M_{hi} .

Proof: ($\subseteq$) Since $\sup N(K \parallel L_m, L, M) \subseteq K \parallel L_m = Q^{-1}(K) \cap L_m$, it suffices to show that $Q(\sup N(K \parallel L_m, L, M)) \subseteq \sup N(K, Q(L), M_{hi})$. By definition, $R = \sup N(K \parallel L_m, L, M)$ is normal w.r.t. L and M . By Lemma 13, under our assumptions, $Q(R) \subseteq K \cap Q(L_m) = K$ is normal w.r.t. $Q(L)$ and M_{hi} . Therefore, $Q(R) = Q(\sup N(K \parallel L_m, L, M)) \subseteq \sup N(K, Q(L), M_{hi})$.

($\supseteq$) To simplify the notation, we henceforth write $\sup N$ for $\sup N(K, Q(L), M_{hi})$. We aim to show that $\sup N \parallel L_m$ is normal with respect to L and M . Suppose $s \in \sup N \parallel L_m$ and $s' \in L$ with $M(s) = M(s')$. It remains to show that $s' \in \sup N \parallel L_m$. Thanks to $s \in \sup N \parallel L_m$, there exists a string $t \in \Sigma^*$ such that $st \in \sup N \parallel L_m$, which implies $Q(st) \in \sup N$. By FMC, we then have $M_{hi}(Q(s't)) = M_{hi}(Q(st)) \in M_{hi}(\sup N)$, which leads to $Q(s')Q(t) \in M_{hi}^{-1}M_{hi}(\sup N)$. Furthermore, by $s' \in L$, it follows that $Q(s') \in M_{hi}^{-1}M_{hi}(\sup N) \cap Q(L)$. The normality of $\sup N$ with respect to $Q(L)$ and M_{hi} therefore implies $Q(s') \in \sup N$. Consequently, $s' \in Q^{-1}(\sup N) \cap L = \sup N \parallel L_m$. By the non-conflictingness assumption, we have $s' \in \sup N \parallel L_m$, which completes the proof. ■

Corollary 15. Given a non-blocking DFA G , let $L = L(G)$. For any prefix-closed high-level specification $K \subseteq Q(L)$, if L is forward mask consistent and strong backward mask consistent with respect to Q , M , and M_{hi} , then $\sup N(K \parallel L, L, M) = \sup N(K, Q(L), M_{hi}) \parallel L$.

Proof: The claim follows from the proof of Theorem

14. In the final step of “ $\supseteq$ ” inclusion, $s' \in Q^{-1}(\sup N) \cap L = \sup N \parallel L$ can be derived without assuming non-conflictingness, as $K \subseteq Q(L)$ is prefix-closed. ■

Theorem 14 indicates that if FMC and SBMC hold, then it is sufficient to compute the high-level supremal normal sublanguage $\sup N(K, Q(L), M_{hi})$, without explicitly computing the low-level supremal normal sublanguage $\sup N(K \parallel L_m, L, M)$.

From a synthesis viewpoint, Propositions 6, 10, and 11 are useful, since they replace the direct verification of FMC and SBMC by structural checks on the mask and the high-level alphabet. The advantage of our abstraction-based framework is in the use of a smaller abstracted plant model compared with the original plant. This is the main benefit of using hierarchical supervisory control. In addition, there is an ideal situation described in Proposition 11, where the abstracted plant has complete observations and the supervisor is computed in polynomial time.

From a computational viewpoint, the sufficient conditions in Propositions 6, 10, and 11 only require checking simple conditions on the mask and the high-level alphabet. The computational complexity for this step is $O(|\Sigma|)$. However, since different events (and strings) may generate the same observation under mask M_{hi} , the induced observation model is typically nondeterministic. Consequently, following the classical exact computation of supremal normal sublanguages in [15], the computation of $\sup N(K, Q(L), M_{hi})$ may require, in the worst case, constructing a nondeterministic finite automaton with at most $n_{hi}n_K$ states and determinizing it into up to $2^{n_{hi}n_K}$ reachable subsets, where n_{hi} and n_K are the numbers of states of G_{hi} and K , respectively. Hence, the computational complexity for this step is $O(2^{n_{hi}n_K})$.

Example 16. Consider an unmanned industrial logistics system consisting of two automated guided vehicles (AGVs) and a robot. Upon a dispatch command, the assigned AGV travels to a feeding buffer, waits for the robot to complete the load-handling operation, and returns. Due to sensing limitations, the two AGVs are indistinguishable under observation.

To model this process, we use generators G_1 and G_2 for the AGVs, and generator G_r for the robot. Let $i \in \{1, 2\}$ denote the index of AGVs. Then, for each $i \in \{1, 2\}$, we use events s_i , e_i , and g_i to stand for respectively start of motion, buffer entry, and buffer departure upon service completion. Fault occurrences while the AGVs are in the buffer are modeled by f_i , and the corresponding repair completions are represented by r_i . The robot reset after each handling cycle is modeled by event b . The component models are shown in Fig. 5.

The global plant is obtained as the product $G = G_1 \parallel G_2 \parallel G_r$. Due to the limited buffer capacity, we impose on G the first-in-first-out specification K , as shown in Fig. 6.

To capture the indistinguishability of the two AGVs, their corresponding events are assigned identical masks, that is, $M(s_i) = s$, $M(e_i) = e$, and $M(g_i) = g$, where $i \in \{1, 2\}$. Furthermore, we have $M(f_i) = M(r_i) = M(b) = \varepsilon$. Moreover, the fault occurrence, repair completion, buffer entry, buffer departure, and robot reset are all treated as uncontrollable events, i.e., $\Sigma_{uc} = \{f_i, r_i, e_i, g_i \mid i \in \{1, 2\}\} \cup \{b\}$.

Since the high-level supervisory control objective focuses on the buffer operation, we choose $\Sigma_{hi} = \{s_i, e_i, g_i \mid i \in \{1, 2\}\}$ as the high-level event set. The corresponding high-level plant over the high-level alphabet Σ_{hi} can then be constructed through a natural projection, as shown in Fig. 7.

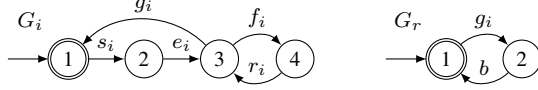


Fig. 5. G_i and G_r modeling the AGVs and robot, respectively.

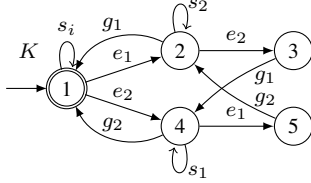


Fig. 6. The specification K .

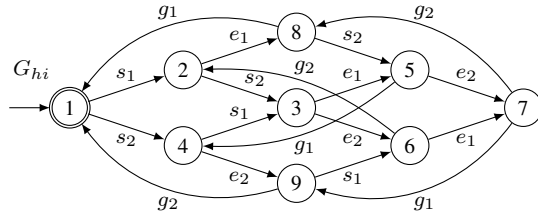


Fig. 7. The high-level plant G_{hi} .

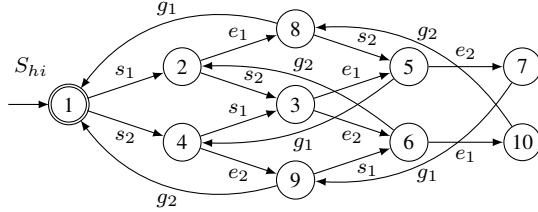


Fig. 8. The high-level supervisor S_{hi} .

It is verified that G satisfies both FMC and SBMC. The FMC follows from Proposition 6, since all events sharing identical masks are either entirely included in or excluded from the high-level alphabet Σ_{hi} . All mask-based observable events are included in the high-level alphabet, i.e., $\Sigma_{om} \subseteq \Sigma_{hi}$, which implies SBMC. Then, by Theorem 14, we have $\sup N(K \parallel L_m, L, M) = \sup N(K, Q(L), M_{hi}) \parallel L_m$.

The constructed high-level supervisor S_{hi} is depicted in Fig. 8, which achieves the language $\sup N(K, Q(L), M_{hi})$. By Theorem 14, this high-level solution can be refined to a non-blocking and maximally permissive supervisor S for the original plant G . Although S_{hi} is synthesized only over the abstracted alphabet Σ_{hi} , its implementation on the original plant G still enforces the first-in-first-out specification K under the mask M . ■

V. HIERARCHICAL COMPUTATION OF SUPREMAL CONTROLLABLE AND NORMAL SUBLANGUAGE

This section extends the mask-based observation framework developed in the previous sections to cope with the presence

of uncontrollable events. Note that the mask affects only the observation map and does not change the controllability status of events, since our abstraction is a natural projection and not a mask. Accordingly, the extension to cover controllability discussed in this section does not need a new notion of controllability under masks.

We first recall the following definition in [5] and [16]. The natural projection $Q : \Sigma^* \rightarrow \Sigma_{hi}^*$ is output control consistent for a language L over Σ , if for all $s \in L$ of the form $s = \sigma_1 \cdots \sigma_k$ or $s = s' \sigma_1 \cdots \sigma_k$, $k \geq 1$, (i) s' ends with an event in Σ_{hi} , (ii) the event $\sigma_i \in \Sigma \setminus \Sigma_{hi}$, for $i = 1, \dots, k-1$, and (iii) the last event $\sigma_k \in \Sigma_{hi}$. Then if σ_k is uncontrollable, all events $\sigma_1, \dots, \sigma_{k-1}$ must also be uncontrollable. We denote high level uncontrollable events by $\Sigma_{uc}^{hi} = \Sigma_{uc} \cap \Sigma_{hi}$.

The projection $Q : \Sigma^* \rightarrow \Sigma_{hi}^*$ has the observer property for a non-blocking plant G over Σ if for all strings $s \in L_m(G)$ such that $Q(s)a \in Q(L_m(G))$ for some $a \in \Sigma_{hi}$, there is a string $u \in \Sigma^*$ such that $su \in L_m(G)$ and $Q(u) = a$.

The following result in [16] shows that the output control consistency and the observer property ensure that the controllability is preserved under projections.

Lemma 17 ([16]). Let L and X be prefix-closed languages over an alphabet Σ with $X \subseteq L$. Suppose that a language $S \subseteq X$ is controllable with respect to L and Σ_{uc} . If the projection $Q : \Sigma^* \rightarrow \Sigma_{hi}^*$ satisfies the observer property and output control consistent for L , then $Q(S)$ is controllable with respect to $P(X)$ and Σ_{uc}^{hi} .

A weaker condition, referred to as local control consistency, is proposed in [10] to ensure the output control consistency. The projection $Q : \Sigma^* \rightarrow \Sigma_{hi}^*$ is local control consistent for a string $s \in L(G)$ if for every $e \in \Sigma_{uc}^{hi}$ with $Q(s)e \in Q(L(G))$, either there is no string $u \in (\Sigma \setminus \Sigma_{hi})^*$ such that $sue \in L(G)$, or there exists a string $u \in (\Sigma_{uc} \setminus \Sigma_{hi})^*$ satisfying $sue \in L(G)$. The projection is local control consistent for a language L if it is true for all strings $s \in L$.

Let G over Σ be a non-blocking plant, K over $\Sigma_{hi} \subseteq \Sigma$ be a specification, and $Q : \Sigma^* \rightarrow \Sigma_{hi}^*$ be the corresponding projection. By Lemma 17, if $\sup C$ is the supremal controllable sublanguage of $K \parallel L_m(G)$ with respect to $L(G)$ and Σ_{uc} , and Q satisfies the observer property and local control consistency for $L(G)$, then the language $Q(\sup C)$ is controllable with respect to $Q(L(G))$ and $\Sigma_{uc} \cap \Sigma_{hi}$.

Now, we combine our results for the normality under masks with the above results from the literature for controllability. We denote by $\sup CN(N, L, \Sigma_{uc}, M)$ the supremal controllable and normal sublanguage of the specification N with respect to the plant language L , the set of uncontrollable events Σ_{uc} and the mask $M : \Sigma^* \rightarrow \Delta^*$.

Theorem 18. Let G be a non-blocking DFA with $L = L(G)$ and $L_m = L_m(G)$. For any high-level specification $K \subseteq Q(L_m)$,

$$\begin{aligned} \sup CN(K \parallel L_m, L, \Sigma_{uc}, M) = \\ \sup CN(K, Q(L), \Sigma_{uc}^{hi}, M_{hi}) \parallel L_m \end{aligned}$$

if 1) $\sup CN(K, Q(L), \Sigma_{uc}^{hi}, M_{hi})$ and L_m are non-conflicting, 2) the projection $Q : \Sigma^* \rightarrow \Sigma_{hi}^*$ has the observer property,

- 3) Q is local (or output) control consistent for L , and
 4) L is FMC and SBMC with respect to Q , M , and M_{hi} .

Proof: The preservation of normality can be shown in the same way as in the proof of Theorem 14. To show the preservation of controllability, we consider two directions.

($\subseteq$) Since $\text{supCN}(K \parallel L_m, L, \Sigma_{uc}, M) \subseteq K \parallel L_m$ is controllable with respect to L and Σ_{uc} , by Lemma 17, $Q(\text{supCN}(K \parallel L_m, L, \Sigma_{uc}, M)) \subseteq K$ is controllable with respect to $Q(L)$ and Σ_{uc}^{hi} . Together with the similar normality argument from Theorem 14 this implies $\text{supCN}(K \parallel L_m, L, \Sigma_{uc}, M) \subseteq \text{supCN}(K, Q(L), \Sigma_{uc}^{hi}, M_{hi})$.

($\supseteq$) The converse inclusion can be done similarly as in [10] by using the fact from [1] that the synchronous composition of the controllable languages is controllable if non-conflictingness is assumed. ■

Corollary 19. Given a non-blocking DFA G over Σ . If $L(G)$ is forward mask consistent, strong backward mask consistent, and local control consistent with respect to Q , M , and M_{hi} , and the projection Q has the observer property, then $\text{supCN}(K \parallel L, L, \Sigma_{uc}, M) = \text{supCN}(K, Q(L), \Sigma_{uc}^{hi}, M_{hi}) \parallel L$ for every prefix-closed high-level specification $K \subseteq Q(L)$, where $L = L(G)$.

Example 20. We revisit the logistics system of Example 16. Recall that the plant is given by $G = G_1 \parallel G_2 \parallel G_r$ and that the first-in-first-out specification is K . As shown in Example 16, the language $L(G)$ satisfies FMC and SBMC.

It remains to verify the other conditions of Theorem 18. By the natural observer verification procedure in [17, 20], it can be verified in polynomial time that Q satisfies the observer property. Thanks to $\Sigma \setminus \Sigma_{hi} \subseteq \Sigma_{uc}$, Q is trivially local control consistent by definition. Therefore, all the conditions of Theorem 18 are satisfied, and it follows that $\text{supCN}(K \parallel L_m, L, \Sigma_{uc}, M) = \text{supCN}(K, Q(L), \Sigma_{uc}^{hi}, M_{hi}) \parallel L_m$. ■

Note that Theorem 14 establishes hierarchical consistency for supremal normal languages, and Theorem 18 extends Theorem 14 to supremal controllable and normal languages. In Theorem 18, since the observer property is assumed to hold, the abstracted plant $Q(L)$ has a smaller state space than the low-level plant [16]. Note that the final worst-case complexity is exponential w.r.t. the smaller (abstracted) plant. A polynomial time complexity can be achieved in the special case of Proposition 11.

By integrating our results, the system can maintain controllability without compromising the preservation of normality. It suffices to compute the supremal controllable and normal sublanguage $\text{supCN}(K, Q(L), \Sigma_{uc}^{hi}, M_{hi})$ to obtain the low-level closed-loop system $L_m(S/G)$. Similarly, the proposed results may provide a theoretical basis for combining mask-based hierarchical synthesis with modular supervisory control frameworks such as [18] and [19], potentially extending them to partially observable modular systems under masks.

VI. CONCLUSION

This article investigates the hierarchical control of DESs under mask-based partial observation. We propose two key

conditions, forward mask consistency and strong backward mask consistency, to ensure the preservation of supremal normal sublanguages between two levels of the hierarchy. It is shown that the non-blocking and maximally permissive normal supervisors at both high and low levels coincide under these consistency conditions. In addition, we propose a stronger condition than forward mask consistency, which effectively supplements its verification. Finally, the presented results also suggest a possible path towards extending the existing modular supervisory control frameworks to the mask-based partially observable setting, which includes modular computation of supervisors resilient to sensor attacks.

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